# Shivayan Chakravarty

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## Education

**Assam Engineering College, Guwahati** **2023 – 2027**
*Bachelor of Technology in Electronics and Telecommunication* *CGPA: 7.87*

## Technical Skills

**Languages:** Java, JavaScript, TypeScript, Python, C, C++, SQL
**Development:** Node.js, Express.js, React.js, Next.js, Tailwind CSS, REST APIs, GraphQL, Socket.IO
**Data & Tools:** MongoDB, PostgreSQL, Redis, RabbitMQ, Docker, Git, GitHub Actions, Cloudinary
**Engineering:** Data Structures and Algorithms, OOP, Microservices, Distributed Systems, RBAC, HLD, LLD, Computer Networks, Operating Systems

## Experience

**Software Engineer Intern** **May 2026 – Jul 2026**
*VrixaaLabs Pvt Ltd* *Hybrid*

- Delivered **3 backend improvements** across a **Node.js/TypeScript microservices** platform, improving request efficiency through centralized throttling and reducing authorization overhead through reusable permission evaluation.
- Architected an **Effective Permission Resolver** aggregating memberships, entity-role assignments, and role-derived permissions; reduced duplicated authorization logic through centralized permission evaluation exposed via **GraphQL**.
- Consolidated **2 duplicated regex implementations** into a shared utility; integrated global rate limiting as well as centralized request throttling across the request pipeline while contributing across **Redis Pub/Sub**, Socket.IO **Redis Adapter**, and MongoDB **connection pooling**.

## Projects

**AI Powered Mock Test Platform - Live | Source Code** **Aug 2025**
*Next.js, Express.js, MongoDB, TypeScript, Redis, RabbitMQ, Docker, Gemini, Groq*

- Engineered **microservices** with 5+ services for scalable and efficient distributed request processing workloads.
- Implemented **asynchronous workflows** with **RabbitMQ** for email delivery, document processing, and AI evaluation, decoupling long-running operations from synchronous API requests and improving system responsiveness.
- Built **AI pipelines** for PDF extraction, text preprocessing, prompt optimization, response validation, and automated model evaluation; implemented **WebSockets** for real-time timed assessment sessions with live progress updates.

**Context Based Social Media Platform - Live | Source Code** **Jan 2026**
*Next.js, Node.js, Express.js, MongoDB, Socket.IO*

- Built a full-stack social platform with **REST APIs**, cookie-based **JWT authentication**, **Google OAuth**, profile onboarding, avatar uploads, and secure session and profile management across user accounts.
- Implemented **real-time one-to-one messaging** with **Socket.IO**, including live message delivery, message deletion events, read states, and real-time notification updates across conversations with persistent message states.
- Developed social graph features: profiles, posts, likes, comments, follows, private accounts, and follow requests.
- Built notifications for follows, likes, comments, and messages with read, delete, and bulk-clear controls.

## Coding Profiles

- Codeforces: Specialist (1419), **150+ problems** solved
- LeetCode: **500+ problems** solved

## Positions of Responsibility

**Project Admin** **May 2026 – Jul 2026**
*Nexus Spring of Code* *Remote*

- Merged **200+ pull requests**, mentored **50+ contributors**, and helped grow the project to **30+ GitHub stars**.

**President** **Jul 2025 – Jun 2026**
*Assam Engineering College* *Guwahati*

- Organized **15+ technical events** and mentored students in **DSA, CP, and software engineering**.